

# LoCoBENCH: A BENCHMARK OF COHERENCE LADDERS FOR EVALUATING LOGICAL COHERENCE MEASURES

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## ABSTRACT

Measuring whether a long-form response *hangs together* — whether the claims it asserts form a consistent argument rather than a bag of individually-checkable facts — requires evaluation data that no existing resource provides. Two documents that differ in coherence almost always differ in topic, length and content as well, so any score difference is unattributable; human coherence ratings supply a target but not a controlled contrast; and nothing supplies a gold *relation graph*, without which a measure’s errors cannot be separated from its relation extractor’s. We introduce **LoCoBench**, a benchmark built from *coherence ladders*: families of five responses over one claim set whose relative coherence ordering is declared in advance, each rung carrying its own machine-readable relation graph derived from a base plan by explicit perturbation operators. A measure is scored not on agreement with a rating but on whether it recovers the declared orderings, and because rungs share a claim set the result is attributable to coherence alone. The corpus targets a documented gap: six of the twelve discourse senses in our annotation scheme — including every exhaustive alternative and every resolved concession — have no gold data in any existing dataset. We specify the generation pipeline as five prompted stages over a 36-topic inventory, all eight prompts reproduced verbatim, with every acceptance threshold centralized and enforced by an independent three-model validation committee that the generator is excluded from; families are admitted whole or not at all, since four of five rungs carry no standalone claim. We report the benchmark’s own validation, including a construction defect we found and fixed whose effect was to make the benchmark silently untestable: when every rung of a family shared the base plan’s relation graph, gold-relation ordering agreement was 2/28, and deriving each rung’s graph from its own operators raised it to 22/28 on identical topics and generator. On the released corpus, measures scored from gold graphs satisfy 78.6% of declared constraints against 35.7–53.6% from mined graphs, and relation extraction tops out at 56.3% coupling-level F1 — locating the field’s bottleneck in extraction rather than scoring.

## 1 INTRODUCTION

Long-form factuality evaluation decomposes a response into atomic claims and verifies each against retrieved evidence (Min et al., 2023; Wei et al., 2024; Song et al., 2024). The paradigm cannot see whether the claims form an argument. A response asserting both that no one was harmed in an incident and that three people died contains one true claim and one false one, and per-claim verification reports exactly that, without registering that the response contradicts itself; a chain of true statements joined by a spurious “therefore” has no unsupported atom in it at all. Xie et al. (2026) quantify the omission: on logic-related samples, SAFE, VeriScore and FactCheck-GPT miss the logical dimension on 60%, 78% and 48% of cases respectively.

Measures of *logical coherence* that address this exist or can be built — from entity grids (Barzilay & Lapata, 2008) and their neural successors (Li & Jurafsky, 2017) through discourse-representation scores (Zhao et al., 2023) to LLM judges (Liu et al., 2023) and probabilistic models over claim-relation graphs. What does not exist is the data to evaluate them on. This paper builds it.

**Why ordinary corpora will not do.** Suppose we wish to know whether a candidate measure tracks coherence. The natural experiment — score a set of documents and correlate against human coherence ratings — has three defects that compound. First, *confounding*: two documents that differ in coherence differ in topic, length, register and claim content as well, so a measure that merely tracks length or claim count can correlate respectably with human ratings for reasons unrelated to coherence. Second, *no ground truth for the mechanism*: a rating tells us a document is incoherent, not *which claims* are in tension, so a measure that gets the right answer for the wrong reason is indistinguishable from one that is right. Third, and most consequential for the class of measures that build an explicit relation graph, *no separation of error sources*: such a measure both extracts relations and scores them, and a rating cannot say which of the two failed.

**What LoCoBench provides instead.** We build families of responses — *ladders* — over a single claim set, where the rungs differ only in a controlled structural respect and their relative ordering is *declared before any measurement*. A conflict ladder holds topic, claims and every unperturbed relation fixed while varying only the conflict structure: a rung with a spurious conflict added, the base, a rung whose conceded tension is resolved, a rung with one conflict removed, and a fully coherent rung. A measure is then scored on the fraction of declared ordering constraints it satisfies. Because the rungs share a claim set, satisfying them is attributable to coherence and nothing else — and because we also declare, per family, the orderings that must *not* hold and one family type in which every rung must score *identically*, the benchmark can detect a measure that is tracking edit count rather than coherence, which is the most likely way such a measure is secretly wrong.

Every rung ships with a machine-readable relation graph: which claims are related, by which discourse sense, in which direction, at what strength, and whether the relation genuinely holds or is a planted error. This is what makes the third defect above tractable. A measure can be run twice — once on its own extracted relations, once on the gold graph — and the difference is its extractor’s contribution, isolated by construction rather than estimated after the fact. In our experiments that difference is large (78.6% against 53.6% of declared constraints), which is itself the most actionable finding the benchmark produces.

**A documented gap in existing annotation.** The relation vocabulary matters here beyond book-keeping. Six of the twelve discourse senses in our scheme — Instantiation, Restatement, Contrast, Concession, Alternative and Disjunction — have no gold data in any existing dataset we are aware of. Two of them are exactly the cases that adjudicative prose turns on: an **Alternative** is a pair of *exhaustive* competing claims of which precisely one must hold, and a **Concession** is a tension the text itself resolves. A measure cannot be evaluated on relations nobody has labelled, and a measure that conflates a mere opposition with an exhaustive one mis-scores every incident investigation and every contested judgment. Creating labelled data for these six is a primary purpose of the corpus, and it drives design decisions throughout: the topic-allocation policy of §7 steers surplus families toward adjudicated subjects precisely because that is where exhaustive alternatives and resolving holdings occur naturally.

Our contributions are as follows.

- **The coherence-ladder construction** (§4): a formal definition, four family types varying different structural dimensions, three classes of declared ordering constraint including predicted *inversions* and a control family that must be flat, and whole-family admission with its justification.
- **A fully specified generation pipeline** (§5): five prompted stages from topic to perturbed rung, six perturbation operators realized as ten parameterized calls, the derivation of each rung’s gold graph from the base plan, and the per-stage failure policy that determines whether a defect is retried, regenerated, or grounds for rejecting the family.
- **Committee validation with centralized thresholds** (§6): three validators run by a model committee that excludes the generator, every threshold in one table, with the empirical calibration of each and an account of two gates that were unsatisfiable as first specified.
- **The item schema and evaluation harness** (§8–§9): arms that hold the scorer fixed and swap only the relation graph, three nested match levels, stratified recall, and declared *non-relations* as a genuine negative set.

- **Validation of the benchmark itself** (§10), including the per-rung relation-graph defect that made an earlier version untestable and the measurements that establish the ladder discriminates between measures at all.
- **All eight prompts verbatim** (Appendix A), extracted programmatically from the released code, with their instruction counts and placeholder contracts.

## 2 RELATED WORK

**Factuality benchmarks and their blind spot.** FactScore (Min et al., 2023) pairs biography prompts with a knowledge source and scores per-atom precision; SAFE (Wei et al., 2024) contributes LongFact-Objects and search-augmented verification with the  $F_1@K$  metric; VeriScore (Song et al., 2024) restricts to verifiable claims. Each supplies claims and evidence, and none supplies inter-claim structure, so none can serve as evaluation data for a coherence measure. Xie et al. (2026) is the closest prior art and the source of our motivating statistic; it constructs data for *logic-related* fact-checking and labels dependencies between claims, but as a classification target rather than as a graded ordering over controlled variants.

**Perturbation-based coherence evaluation.** Evaluating coherence models against synthetic perturbations is long-established: entity-grid work (Barzilay & Lapata, 2008) discriminates a document from its sentence permutations, and neural coherence models (Li & Jurafsky, 2017) extend the practice with insertion and replacement perturbations. LoCoBench inherits this methodology and changes one thing. There, the perturbation is an operation on *text* whose effect on coherence is assumed; here it is an operation on a *declared relation graph*, whose effect on the graph is computed and recorded per rung. Consequently the benchmark can state not only that rung  $k + 1$  should score higher than rung  $k$  but also *why*, which relation changed, and — for the ordering-only and control operators — that certain rungs must score *identically* because no factor changed at all. Sentence shuffling alone cannot express that distinction.

**Discourse annotation as precedent.** Our sense inventory is grounded in PDTB 3.0 (Prasad et al., 2019) and RST (Mann & Thompson, 1988), and the debt is methodological as well as terminological: both establish that discourse relations can be annotated reliably from surface cues, which is what makes V1 of §6 a reasonable gate to impose. The departure is that PDTB annotates naturally-occurring text, where the relation inventory’s distribution is whatever the corpus happens to contain — which is precisely why six of our senses have no gold data. A generated corpus can *require* the rare relations, and ours does: every plan must contain at least one Alternative, one Disjunction, one Restatement, one Concession and one temporal ordering.

**LLM-generated corpora and their validity risks.** Using a language model to generate evaluation data is now common, and the known risks are circularity (the generator’s biases become the benchmark’s ground truth), leakage (artifacts revealing the generation procedure), and unverifiable labels. We address these structurally rather than by assertion: the validating committee excludes the generating model (§6), one validator’s sole purpose is to detect leakage spans, and the V1 gate requires an independent model to *recover* the planned relations from the prose before the item is admitted, so a plan the text fails to realize is rejected rather than shipped with aspirational labels. §11 states plainly what this does not fix.

## 3 THE ANNOTATION SCHEME

An item’s gold annotation is a graph over its claims, and the edge labels come from a two-level scheme: an interpretable *discourse sense* that an annotator can read off surface cues, and an inferential *coupling* that says what the relation asserts about the joint truth of its endpoints. The two levels are connected by a deterministic map, and that map — not the annotator, and not the generator — is the authority on what a sense means.

Table 1: The compile map  $\mathcal{C}$  of Def. 2. *Ordering-only* senses record assertion order but carry **no truth coupling**, so they produce no factor in any scorer that consumes this scheme — which is what makes the ordering-only perturbation of §5 a meaning-preserving edit whose score must not move. **Concession** marks a conflict the text may itself resolve. The nine senses marked  $\star$  are the combinations the corpus actually labels; **bold** marks the six senses for which we are aware of no gold data in any existing dataset.

| Level-2 sense                        | $\mathcal{C}$ coupling | prior   | dir. | flag  | gloss                                  |
|--------------------------------------|------------------------|---------|------|-------|----------------------------------------|
| Cause-Effect $\star$                 | ENTAILMENT             | $\perp$ | yes  |       | source brings target about             |
| Effect-Cause $\star$                 | ENTAILMENT             | $\perp$ | yes  |       | source is the effect, target the cause |
| Evidence $\star$                     | ENTAILMENT             | $\perp$ | yes  |       | source is evidence for target          |
| Condition                            | ENTAILMENT             | $\perp$ | yes  |       | source conditions target               |
| <b>Instantiation</b>                 | ENTAILMENT             | $\perp$ | yes  |       | target instantiates source             |
| <b>Restatement<math>\star</math></b> | EQUIVALENCE            | 0.90    | no   |       | definitional identity                  |
| <b>Contrast<math>\star</math></b>    | CONTRADICTION          | $\perp$ | no   |       | opposed, <i>not</i> exhaustive         |
| <b>Concession<math>\star</math></b>  | CONTRADICTION          | $\perp$ | yes  | conc. | a tension the text resolves            |
| <b>Alternative<math>\star</math></b> | EXCLUSIVE              | $\perp$ | no   |       | exhaustive: exactly one holds          |
| <b>Disjunction<math>\star</math></b> | CO_NECESSITY           | $\perp$ | no   |       | at least one holds                     |
| Precedence $\star$                   | NONE                   | —       | yes  | ord.  | temporal order only                    |
| Succession                           | NONE                   | —       | yes  | ord.  | temporal order only                    |

### 3.1 LEVEL ONE: INFERENTIAL COUPLINGS

**Definition 1** (Inferential couplings). A coupling  $\tau$  is specified by the set  $V(\tau) \subseteq \{0, 1\}^2$  of joint truth assignments  $(a_s, a_t)$  of its source and target claims that it penalizes — the configurations at which the relation is violated:

| Coupling      | Reading                                            | Directed | Penalized set $V(\tau)$ |
|---------------|----------------------------------------------------|----------|-------------------------|
| ENTAILMENT    | $a_s \Rightarrow a_t$                              | yes      | $\{(1, 0)\}$            |
| CONTRADICTION | $a_s \Rightarrow \neg a_t$                         | yes      | $\{(1, 1)\}$            |
| EQUIVALENCE   | $a_s \Leftrightarrow a_t$                          | no       | $\{(0, 1), (1, 0)\}$    |
| EXCLUSIVE     | <i>exactly one of <math>a_s, a_t</math> holds</i>  | no       | $\{(0, 0), (1, 1)\}$    |
| CO_NECESSITY  | <i>at least one of <math>a_s, a_t</math> holds</i> | no       | $\{(0, 0)\}$            |
| NONE          | <i>no truth coupling</i>                           | —        | $\emptyset$             |

The two couplings whose penalized set contains the both-true world,  $\{\text{CONTRADICTION}, \text{EXCLUSIVE}\}$ , are the conflict couplings.

The distinction between CONTRADICTION and EXCLUSIVE is the one the scheme exists to draw, and it is the exhaustiveness question. CONTRADICTION forbids only the both-true world: two claims that cannot both hold but might both fail. EXCLUSIVE additionally forbids the both-false world, which is what an exhaustive pair asserts — “no one was harmed” and “three people died” can neither both hold nor both fail. CO\_NECESSITY is the dual, forbidding only the both-false world, which is what a disjunction asserts. An annotation scheme that offers only CONTRADICTION cannot express either, and any measure evaluated against such a scheme is untested on precisely the relations that adjudicated disputes turn on.

### 3.2 LEVEL TWO: DISCOURSE SENSES, AND THE COMPILE MAP

**Definition 2** (Compile map). The compile map  $\mathcal{C}$  assigns to each discourse sense a coupling, a strength prior ( $\perp$  meaning “use the estimated value”), and the flags directed, concession and ordering-only. It is total and deterministic, and is given in Table 1.

Five senses collapse onto ENTAILMENT, which is a deliberate loss of information at the inferential level: whether one claim causes another or merely evidences it makes no difference to what the pair asserts about joint truth. The sense is retained alongside the coupling so that error analysis can distinguish “related, but mislabelled” from “not related”, and so that the corpus remains useful to work that cares about the discourse layer for its own sake.

Table 2: The four family types. Each holds the claim set, the topic and every unperturbed relation fixed, varying one structural dimension. CONTROL is the false-positive check: every one of its edits is meaning-preserving and adds no factor, so a measure showing any monotone trend across it is tracking edit count rather than coherence. Counts are the target allocation at a corpus of 120 families.

| Type     | Dimension varied             | Rungs, $\rho_0 \rightarrow \rho_4$                                         | Required pattern                                 | $n$ |
|----------|------------------------------|----------------------------------------------------------------------------|--------------------------------------------------|-----|
| CONFLICT | conflict structure           | worse, base, concession_resolved, fix_one_conflict, coherent               | strictly increasing                              | 55  |
| CHAIN    | entailment-chain integrity   | break_3, break_2, break_1, base, coherent                                  | strictly increasing                              | 25  |
| ORDER    | assertion order              | shuffle_full, shuffle_block, shuffle_adjacent, base, ordering_only         | increasing, then <i>exactly flat</i> at $\rho_4$ | 25  |
| CONTROL  | nothing (meaning-preserving) | ordering_only, direction_reversal, base, direction_reversal, ordering_only | <b>flat throughout</b>                           | 15  |

**The compile map is the authority, and it is enforced.** A gold relation carries both a sense and a coupling, which invites inconsistency. The schema validator rejects any item in which coupling  $\neq \mathcal{C}(\text{sense})$ , and likewise re-derives directedness, ordering-only status and exhaustiveness from the sense rather than trusting the generator’s assertion of them. This matters for a generated corpus in a way it would not for a hand-annotated one: an LLM asked for both a sense and a coupling will occasionally emit a pair that no reading of the scheme licenses, and silently keeping such an edge would place a factor in the graph that the sense does not justify. Nine (sense, coupling) combinations occur in the corpus and the mapping among them is a bijection, so an edge outside that set cannot be scored against gold at all.

#### 4 COHERENCE LADDERS

**Definition 3** (Coherence ladder). A ladder is a tuple  $(\mathcal{P}, \langle \rho_0, \dots, \rho_4 \rangle, \Gamma)$  where  $\mathcal{P}$  is a base relation plan — a selected claim set with a declared relation graph over it — each rung  $\rho_k$  is a sequence of perturbation operator calls applied to  $\mathcal{P}$ , yielding a response text together with its own gold relation graph, and  $\Gamma$  is a set of constraints on the relative ordering of the rungs’ scores. A ladder is admitted only if all five rungs pass every validation gate of §6.

**Whole-family admission.** A ladder is admitted entire or not at all. This follows from what the rungs are for: four of the five carry no standalone claim, only a claim relative to their neighbours, so a ladder missing a rung is not a partial success but a broken comparison. The gate therefore sits at family granularity, and a family whose fourth rung fails V3 is rejected together with the four rungs that passed.

Table 2 gives the four types. CONFLICT is the reference ladder and the one the released corpus instantiates: only the conflict structure varies, which is what makes an ordering attributable to coherence rather than to content. CHAIN tests a different failure mode, breaking  $N$  links of an entailment chain and repairing them, where the defect is a *missing* relation rather than a present conflict — a distinction that matters because a measure built on contradiction detection alone will do well on the first ladder and fail the second. ORDER varies assertion order at three granularities and then adds a rung that must be flat.

CONTROL deserves emphasis because it is the type most benchmarks of this shape omit. Every rung applies a meaning-preserving edit — exchanging a **Precedence** for a **Succession**, which Table 1 maps to no coupling at all, or reversing the direction of a symmetric relation — so the required pattern is a flat line. A measure that trends monotonically across CONTROL has demonstrated that it responds to *having been edited*, which is a false positive no increasing-order ladder can detect. We regard this

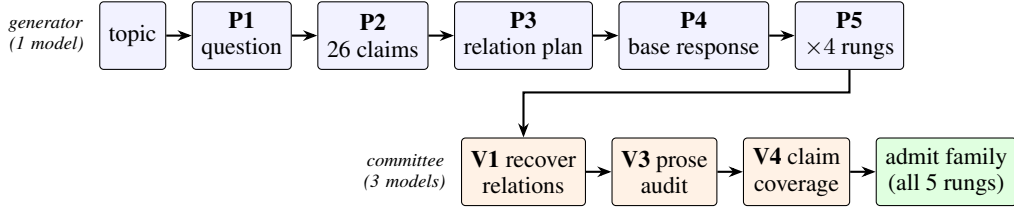


Figure 1: The pipeline. Five generation stages produce a family of five rungs from one topic; three validators, run by a committee that excludes the generating model, gate admission. A family is admitted whole or rejected whole (Def. 3).

as the single most important experiment the protocol specifies, and we are candid in §11 that the released corpus does not yet contain it.

#### 4.1 ORDERING CONSTRAINTS

A family declares its constraints in three classes, each asserted per readout of the measure under test.

**C1, strict increase.** Consecutive rungs must be strictly ordered. Listed per readout per pair, so a family can assert that one readout must increase between two rungs while another is unconstrained there — which is needed because different readouts of the same model legitimately have different sensitivities.

**C2, predicted inversion or invariance.** A constraint that a score must *decrease*, or must be *exactly invariant*. This class is what distinguishes a declared-ordering benchmark from a monotonicity test, and it has two sources. Some measures are expected to dip when a conceded tension is resolved, because softening a conflict raises the probability of the very configuration the conflict forbids; asserting that dip positively turns a known peculiarity into a falsifiable prediction rather than an excuse offered after the fact. And an ordering-only edit *must* be exactly flat, since Table 1 gives it no coupling: here C2 asserts an equality, and a measure that moves at all has been caught reading assertion order directly.

**C3, endpoint separation.** The extreme rungs must separate, for all readouts. This is the robust headline claim, insensitive to the fine structure of the intermediate rungs, and the one we would report if only one constraint class could be.

A measure is scored on the fraction of declared constraints satisfied, with 14 constraints per family in the CONFLICT type.

## 5 THE GENERATION PIPELINE

Generation is a five-stage prompted pipeline followed by validation, and the stages are separated so that a defect can be attributed to one of them and repaired there. Each stage’s prompt is fixed, versioned, and reproduced verbatim in Appendix A; the instruction count and placeholder set of each is declared in a specification table and asserted by a test, so a prompt cannot drift from the document that describes it without the test failing.

### 5.1 P1: TOPIC TO QUESTION

Given a topic from the inventory of §7, emit one open-ended question. Three of the nine instructions are doing real work. The question must *elicit* diverse logical structure, and the prompt enumerates the target relations so that the question is chosen with the downstream plan in mind rather than for topical interest. It must prefer adjudicative domains — incident investigations, litigation, diagnostic reasoning, attribution disputes, competing scientific hypotheses — because these are the settings in which exhaustive alternatives and resolving holdings arise naturally, and those are the scarce relations of Table 1. And it must *not* contain the words “cause”, “condition”, “alternative” or “contrast”, so that the relations the answer draws are not primed lexically by the question: without this constraint

the generator tends to echo the question’s vocabulary and the resulting relations are an artifact of the prompt rather than of the subject.

## 5.2 P2: QUESTION TO TAGGED CLAIMS

Emit exactly 26 atomic claims, each a single declarative sentence with no connective joining it to another — relations are added at the next stage, not here, so that the claim set and the relation graph are independently controlled. The composition is specified rather than left to chance: 14 factually correct claims; 4 plausible-but-false claims; an *exhaustive alternative* pair, of which exactly one must be true; a *disjunctive* pair, of which at least one must be true but both may be; an *equivalent* pair asserting one fact in different words; and two *holding* claims in which an authority reaches a determination settling a dispute, marked by adjudicating language. Each claim carries a tag recording which of these roles it fills.

Two details prevent the composition from being defeated. The disjunctive pair must be phrased as states of the world rather than as findings — “the alloy was substituted” and “the batch was mislabelled”, not “assay A flagged it” and “assay B flagged it” — because a pair of *reports* is trivially compatible and yields no real disjunction. And the false claims must not be signalled in their wording: no hedging with “allegedly” or “possibly”, since a response that flags its own false content is not the artifact the benchmark needs.

## 5.3 P3: CLAIMS TO RELATION PLAN

The planning stage, and the most constrained. Select 14–16 of the claims and renumber them to consecutive assertion positions, then plan exactly ten relations over distinct claim pairs, each carrying a sense from Table 1, an intended strength band (strong, moderate, or weak), and a *validity* label. Four requirements shape the output.

*At least two false claims must be selected*, so the response asserts false content alongside true. Without this quota the planner selected none and the shipped corpus was entirely factual — which would make coherence and factuality perfectly correlated in the data and so unable to distinguish a measure of one from a measure of the other. The quota is two rather than four because P2 writes four false claims among 26 and P3 selects about 15, so roughly 2.3 are expected under indifferent sampling: the quota sits just below the natural rate, and demanding all four would force the planner’s hand on four of fifteen slots and collide with the required-sense quota below.

*Five senses are required*: at least one *Alternative*, one *Disjunction*, one *Restatement*, one *Concession*, and one temporal ordering. These are the scarce facets of §1, and requiring them is what makes the corpus supply gold data for relations existing corpora lack.

*Exactly six relations valid and four invalid*, with each invalid relation carrying an error kind — *wrong\_sense*, *wrong\_direction*, *false\_endpoint* or *spurious*. The count is stated as a count rather than a proportion because a planner asked for “about 60% valid” computes a percentage and rounds unpredictably. Planted invalid relations are what make the corpus a test of *precision* as well as recall: a relation extractor that recovers every gold edge including the deliberately wrong ones has not demonstrated good judgement.

*Four to six declared non-relations*: pairs of selected claims, within four positions of each other, between which the response must draw no dependence. These are the distractors, and they are the corpus’s most useful and least obvious feature. Precision measured against all unlabelled pairs is close to meaningless, since most unlabelled pairs are simply unannotated rather than known to be unrelated; a violated *declared* non-relation, by contrast, is a genuine false positive against a real negative set. The two lists are required to be disjoint.

Relations are also kept local by default — preferring pairs within four positions — with long-range pairs permitted where the two claims share a named entity. This is not a claim about discourse in general but a deliberate bias toward graphs that a local reader can recover, so that a failure of the V1 gate indicates a defective realization rather than an unreasonable demand.

Table 3: The six perturbation operators and the ten parameterized calls realizing them. Each call also has a declared effect on the rung’s *gold relation graph*, so a rung’s edges are the base edges with its own perturbations applied (§5.6). The two O6 calls are factor-invariant by design: they are what the ORDER and CONTROL ladders exist to check.

| Op               | Calls                                                      | Effect on the text                                                                                                                                               |
|------------------|------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| O1 relabel       | wrong_sense,<br>direction_reversal,<br>exhaustiveness_flip | rewrite the connective so the text draws a different sense, reverses the relation, or flips Contrast $\leftrightarrow$ Alternative                               |
| O2 add conflict  | inject_contradiction,<br>spurious_relation                 | add one sentence directly contradicting a named claim, at least two sentences later; or assert a link between two claims the response kept separate              |
| O3 break chain   | break_chain                                                | remove or negate the link realizing an edge so the chain no longer connects, keeping both endpoints                                                              |
| O4 drop relation | drop_relation                                              | remove the connective realizing an edge, leaving the two claims as adjacent assertions                                                                           |
| O5 resolve       | add_resolution,<br>remove_resolution                       | add or delete the holding sentence that settles a conceded tension                                                                                               |
| O6 reorder       | shuffle_order,<br>ordering_only                            | permute sentences at full, block or adjacent granularity; or exchange Precedence $\leftrightarrow$ Succession preserving chronology — <i>no coupling changes</i> |

#### 5.4 P4: PLAN TO RESPONSE

Write prose realizing the plan exactly: assert every planned claim once, in position order, and realize every planned relation so that a careful reader could recover it from the text alone. This stage carries the pipeline’s hardest instruction, because two requirements pull against each other — the text must be recoverable as a relation graph, and it must read as publishable expert prose rather than as an annotation walked sentence by sentence.

Several instructions exist to resolve that tension. Referring back to an already-asserted claim must use a pronoun or a short noun phrase named after *the thing the claim is about*, not after its status as a finding: “the primitive carpal morphology”, not “the analysis’s flagging of primitive traits” — the latter is precisely the register that reads as stitched-together annotation. Quantifiers, determiners, numbers, dates, polarity and modals are declared to be *content* rather than wording, so “all tools were crafted by a single culture” may be reworded but not weakened to “the tools were”, which would silently change the claim being asserted and defeat V4. Exhaustive and disjunctive pairs must make their logical strength *explicit*, since a bare “X, or Y” leaves a reader unable to tell whether both could hold — which is the only thing separating Alternative from Disjunction. Temporal orderings must be stated within a single sentence joining both claims, because an ordering a reader cannot see is the same as no ordering. Non-relations must be separated by distance and paragraphing rather than by an explicit marker: announcing “in an unrelated matter” would both leak the plan and, ironically, constitute drawing attention to a relation.

#### 5.5 P5: RESPONSE TO RUNG

Apply exactly one perturbation to the base response, changing the minimum number of words and preserving everything else verbatim. Table 3 gives the ten calls, grouped by the six operators they realize.

The stage also emits a structured diff naming the operator, its target, the sentences changed, and the relations added, removed or relabelled. Length drift is capped at 15%, except for the three calls that add or remove a sentence by definition.

#### 5.6 DERIVING EACH RUNG’S GOLD RELATION GRAPH

A rung’s gold graph is the base plan’s graph with that rung’s operator calls applied. Three mechanisms make this well defined.



First, *target assignment*. Each call in a rung is assigned a distinct eligible edge, so a rung composing two `drop_relation` calls drops two *different* edges. Eligibility is call-dependent: a `drop_relation` in a conflict ladder targets an unresolved conflict edge, an `add_resolution` targets a resolvable one. A CONFLICT plan therefore needs three unresolved conflict edges to support its full ladder — one for the resolution and two for the drops — and a plan with fewer is rejected at generation time rather than silently emitting a rung that duplicates its neighbour.

Second, *derived-field re-derivation*. When a call changes an edge’s sense, the coupling, directedness, ordering-only status and exhaustiveness are re-derived from the new sense through  $\mathcal{C}$  rather than carried over, so a transformed edge cannot contradict the compile map.

Third, an *adjacency gate*. A family is rejected when two *adjacent* rungs build the same edge set while the higher rung’s calls claim a change. Adjacency rather than parentage is the right comparison: every CONFLICT rung declares the base as its parent, so a parent-only check passes a rung that duplicates the rung immediately below it, which is exactly the defect that made an earlier version of the corpus untestable (§10.1). The two factor-invariant calls of O6 are exempt, since they are *intended* to leave the edge set unchanged.

## 5.7 THE FAILURE POLICY

What happens when a stage’s output is defective is a specification decision, not an implementation detail, and the policy is per stage with four outcomes. A **retry** handles a transient parse failure where the inputs are sound: call again. Because a byte-identical prompt re-sent to a temperature-zero backend cannot produce a different parse, retries are issued at successively higher sampling temperatures, which is what makes a retry budget worth more than one. A **regenerate** handles an artifact that is wrong while its inputs are fine: redo that artifact. A **drop edge** applies to V1 alone — an unrecovered relation must never be retained as gold, since gold that the prose does not realize is exactly the defect the gate exists to catch. A **reject** applies when the family cannot carry its ranking claim, and stores the family with its reason rather than discarding it, so that rejection rates are auditable and the rejected set is available for analysis.

One further guard concerns output length. P4’s floor is 500 words with a target of 550–650, which exceeds the default completion-token grant of some gateways; a model that hits the cap returns prose stopped mid-sentence, which every parser rejects — but for the wrong stated reason, reporting a malformed response where the actual fault is a truncated generation. The ceiling is therefore raised explicitly rather than left to the backend default.

## 6 VALIDATION

An item is admitted only if an independent committee certifies it. Three properties of the arrangement matter more than the individual gates.

**The committee excludes the generator.** Validation is run by a panel of models, and the model that generated the family is removed from its own panel before voting. Without this, the dominant failure of an LLM-generated benchmark — the generator certifying its own idiosyncrasies as ground truth — is structurally unaddressed. Each gate is decided by *majority* of the remaining panel, and a facet is counted as failed only when a majority objects, so a single unusually strict auditor cannot reject a sound family.

**Every threshold lives in one place.** All acceptance thresholds are defined in a single table (Table 4) and nowhere else. This is a reproducibility property: a benchmark whose thresholds are scattered across the code that applies them cannot be replicated from its description, and cannot be re-run at a different operating point without a search for the constants.

**Generation is rejected, never repaired.** A family that fails a gate is stored with its reason and discarded from the corpus. No stage attempts to patch a defective artifact into an acceptable one, because a repaired item’s gold labels no longer describe a process we can state.

Table 4: Every acceptance threshold. V1 rates are computed over the *valid* planned relations only. Rules in the rightmost column are the committee decision rule for that gate.

| Gate   | Quantity                                             | Threshold          | Rule     |
|--------|------------------------------------------------------|--------------------|----------|
| V1     | planned relations recovered with correct coupling    | $\geq 0.80$        | majority |
| V1     | ... and with correct sense                           | $\geq 0.70$        | majority |
| V3     | fluency, formality, organization (each, of 5)        | $\geq 4$           | majority |
| V3     | leakage and hedging span lists                       | must be empty      | majority |
| V4     | claim coverage over gating statuses                  | $= 1.00$           | majority |
| V4     | gating statuses                                      | missing, merged    | —        |
| P3     | claims selected                                      | 14–16              | —        |
| P3     | relations planned                                    | 8–12               | —        |
| P3     | declared non-relations                               | 4–6                | —        |
| P3     | fraction of relations valid                          | $0.55 \pm 0.15$    | —        |
| P3     | false claims selected                                | $\geq 2$           | —        |
| P3     | preferred relation locality (recorded, not enforced) | $\leq 4$ positions | —        |
| P5     | length drift                                         | $\leq 15\%$        | —        |
| Corpus | families per topic                                   | $\geq 3$           | —        |

## 6.1 V1: RELATION RECOVERY — THE LOAD-BEARING GATE

Given the response and its claims, an independent model is asked to recover the relations the response *itself* draws, emitting for each ordered pair a sense, a coupling, a strength band, and a resolved flag. Admission requires that at least 0.80 of the *valid* planned relations come back with the correct coupling and at least 0.70 with the correct sense.

This is the gate that makes the corpus’s labels mean something. Everything else in the pipeline produces a plan; V1 certifies that the *prose realizes it*, so the gold graph describes the response rather than the generator’s intention. Without it the corpus would ship aspirational annotations: a relation the planner intended and the writer failed to express would still appear in the gold graph, and a relation extractor would then be penalized for correctly not finding it.

Two design decisions inside V1 are worth stating because both were arrived at by measuring a failure.

*The rates are computed over valid relations only.* Planted-invalid relations are deliberately wrong, so requiring an independent model to recover them would gate admission on the model’s willingness to assert relations that do not hold — rewarding exactly the over-assertion the corpus is built to detect. With the mandated six valid relations the denominator is six, making these integer gates: 0.80 needs five of six, and so does 0.70, since four of six falls below both. One relation therefore decides admission. It follows that the coupling threshold must not be raised above 0.834, at which point it silently becomes unanimity-of-six — a fresh unsatisfiable gate of exactly the species that scoring the planted errors used to be. Measured across seven raters and two items, valid-only recovery rates ranged 0.833–1.00, so the gates clear with one relation to spare.

*Recovery is graded at three levels, not one.* A recovered relation is *identified* by its unordered claim pair, independently of its sense; direction, coupling and sense are then graded separately. Deriving the identity key from the sense instead — ordered keys for directed senses, unordered for symmetric ones — puts a planned and a recovered relation over the *same two claims* into different key spaces whenever their senses disagree about directedness, so the pair becomes unmatchable and scores zero on sense *and* zero on coupling. Measured live: one model recovered a planned **Concession**(7, 12) as **Contrast**(7, 12) — same endpoints, and both senses compile to CONTRADICTION — and earned nothing, because the two senses keyed differently. That counts “recovered with the wrong sense” as “not recovered”, which is precisely the distinction V1 exists to measure. Direction is not discarded by the fix, it moves: endpoint order is compared separately and credit withheld when a directed planned relation was recovered reversed, so a wrong-direction error is visible as a matched pair that failed on direction rather than as an absence.

Table 5: The topic inventory: 36 topics across four domains. Surplus families beyond the per-topic floor are allocated in priority order to the five **bold** topics, which are the adjudicated subjects where exhaustive alternatives and resolving holdings occur naturally.

| Domain                   | Topics                                                                                                                                                                       | <i>n</i> |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Natural Sciences         | Astronomy, Biology, Botany, Chemistry, Geology, Human Biology, <b>Medicine</b> , Meteorology, Oceanography, Physics, Zoology                                                 | 11       |
| Humanities               | Archaeology, Art, Culture, <b>Ethics</b> , Gender Studies, History, Linguistics, Literature, Philosophy, Religion                                                            | 10       |
| Engineering & Technology | Artificial Intelligence, <b>Civil Engineering</b> , Computer Science, Cybersecurity, Electrical Engineering, Mechanical Engineering, Renewable Energy Technologies, Robotics | 8        |
| Social Sciences          | Anthropology, Economics, Education, <b>Law</b> , <b>Political Science</b> , Psychology, Sociology                                                                            | 7        |

## 6.2 V3: PROSE AUDIT

The response is scored for fluency, formality and organization, each on 1–5 with a floor of 4, and three span lists are collected: *leakage*, *hedging* and *artifacts*. Leakage and hedging must be empty; artifacts are recorded but do not gate.

Leakage is the interesting one, and its definition is deliberately narrow. It covers naming a relation type or sense, naming the plan or its labels, referring to “claims” or “atoms” as the objects the text was assembled from, and any bracketed marker or numeric index. It explicitly does *not* cover the connectives that *realize* relations — “either X or Y”, “at least one of these must hold”, “although X, Y”, “which postdates” — because P4 is required to use them, and a coupling over two claims’ joint truth cannot be expressed without one. An earlier, broader reading of leakage rejected correctly-written responses for containing the very markers the pipeline demands. Every reported span must be quoted verbatim from the response, which makes a hallucinated objection checkable and therefore discardable.

Hedging is defined by a *closed* list of seven words. The reason is the same: uncertainty a writer is required to express is not hedging, so “at least one of X or Y, perhaps both” states a relation’s logical strength rather than doubt about content, and an open-ended reading flags it.

## 6.3 V4: CLAIM COVERAGE

For each claim, an independent model decides whether the response asserts it, returning one of **asserted**, **altered**, **missing** or **merged** with a quoted span. Coverage must be perfect over the *gating* statuses, which are **missing** and **merged**.

**altered** is deliberately non-gating, and this is a calibration decision rather than a lax one. It is the judgment most sensitive to the surface rewording P4 explicitly licenses, and three of five live V4 rejections were a single claim called **altered** where the rewording was permitted. **missing** and **merged** are structural losses — the claim is not separately assertable any more — and those are what coverage is for. The prompt also states explicitly that two claims joined by a connective are *not* merged, since “either X or Y” asserts both and each remains independently true or false; without that clarification the validator counted every realized **Alternative** as a merge, which would have rejected every family containing the corpus’s scarcest relation.

## 7 CORPUS COMPOSITION

Table 5 gives the inventory. Allocation is floor-then-surplus: every topic contributes at least three families without exception, and the remainder is distributed round-robin over five priority topics. The floor exists so that no measure can be tuned to a topic distribution; the priority list exists because the scarce relations of Table 1 are not uniformly available across subjects. Litigation, medicine and civil engineering produce exhaustive alternatives and authoritative holdings as a matter of course, whereas botany and art history mostly do not, so a uniform allocation would leave **Alternative** and **Concession** underrepresented exactly where the corpus is supposed to supply them.

Five topics are named differently in the source whose topic list ours extends; the alias mapping is recorded with the corpus so that a topic list can be diffed against either source rather than silently diverging.

**The released corpus.** What we release and report on here is smaller than the design target: two CONFLICT families, ten items, on the canonical topics Anthropology and Archaeology, sixteen claims per item, generated by `claude-opus-5` with a validation committee of `gpt-5.6`, `gemma-3.1-pro` and `claude-sonnet-5`. The CHAIN, ORDER and CONTROL ladders are specified, implemented and gate-checked but not generated at scale. §11 states the consequences; we report the released corpus’s numbers as measurements of a pilot rather than of the design.

## 8 ITEM SCHEMA AND RELEASE FORMAT

Table 6 gives the schema. The release is four files per corpus: `items.jsonl`, one item per line; `families.json`, the manifest carrying each family’s declared ordering constraints; `state.json`, the generation log; and a `rejected/` directory holding every family that failed a gate together with the gate, the threshold, the observed value and — where the failure was a parse — the raw model output. The rejected set is part of the release rather than an artifact of development: reported admission rates are only auditable if the denominator is available.

Two schema decisions carry weight. Derived fields are re-computed rather than trusted, so the corpus cannot contain an edge whose coupling contradicts its sense (§3). And `ordering_only` is explicit, so a consumer can partition the gold graph into the edges that produce factors and the edges that record order alone — which is what allows the ORDER and CONTROL predictions of §4 to be checked mechanically.

## 9 THE EVALUATION HARNESS

### 9.1 ARMS: HOLDING THE SCORER FIXED

An *arm* determines where the relation graph comes from while holding everything else identical — the claims, the priors, and the scoring procedure. Three arms are defined. GOLD takes each item’s own labelled relations. GOLD\_VALID takes only those labelled `valid`, which quantifies what the planted errors cost at a fixed rung. A MINED arm, written `mined:{model}:{policy}`, runs a relation extractor on the response text.

This is the harness’s central device. Because the scorer is byte-identical across arms, any shortfall of a mined arm below the gold arm is attributable to relation extraction and not to scoring, and the two error sources need never be disentangled after the fact. The gold arm additionally requires no LLM at all, so it is deterministic and cheap, which makes it usable as a regression test on the corpus itself.

### 9.2 WHAT A SCORER MUST EXPOSE

The harness is agnostic about the measure under test, requiring only that it map a relation graph over claims to one or more scalar readouts. For concreteness we describe the family of measures we use to validate the corpus, which builds a Markov random field over claim truth variables: one binary variable per claim with unary factor  $[1 - \pi_i, \pi_i]$ , and one pairwise factor per relation determined by its coupling and weight, penalizing the assignments  $V(\tau)$  of Def. 1. Four readouts over the resulting joint are used: the mean posterior marginal  $\frac{1}{n} \sum_i \mathbb{P}(a_i=1)$ ; a two-term consistency probability combining the probability that no conflict is jointly active with the confidence-weighted mass of actively-upheld support relations; the marginal of an added coherence node tied to relation satisfaction; and the partition function normalized between a conflict-free ceiling and the base network’s most-probable world. Gold graphs are scored with  $\pi_i = 0.9$  for claims the item marks factual and 0.1 otherwise, so that the planted-false claims enter the graph already doubted.

The relevant point for the benchmark is that these four readouts have *different* declared expectations under C2. One of them is expected to dip when a conceded tension is resolved — softening a conflict raises the probability of the configuration the conflict forbids, so a readout that measures whether a conflict is active must register the softened conflict as more often active — and the manifest asserts

Table 6: The item schema. Every field is machine-readable and validated on load; fields marked *derived* are re-computed from the sense through  $\mathcal{C}$  rather than trusted as written (§3).

| Field                                | Type           | Meaning                                                                            |
|--------------------------------------|----------------|------------------------------------------------------------------------------------|
| <i>Item level</i>                    |                |                                                                                    |
| id, name                             | string         | stable identifier and human-readable label                                         |
| source                               | string         | generating model                                                                   |
| response                             | string         | the prose                                                                          |
| num_atoms                            | int            | claim count                                                                        |
| notes                                | string         | free-text provenance note                                                          |
| <i>Per claim (atoms)</i>             |                |                                                                                    |
| id, text                             | string         | claim identifier and its exact text                                                |
| label                                | string         | P2 tag: correct, incorrect, alt-pair, disj-pair, equiv-pair, holding               |
| factual                              | bool           | whether the claim is true — <i>false</i> for planted-false claims                  |
| role                                 | string         | structural role in the plan                                                        |
| pos                                  | int            | 1-based assertion position                                                         |
| <i>Per relation (relations)</i>      |                |                                                                                    |
| id                                   | string         | edge identifier, referenced by perturbation targets                                |
| source_id, target_id                 | string         | endpoint claims, ordered                                                           |
| level2_sense                         | string         | one of the twelve senses                                                           |
| level1_coupling                      | string         | <i>derived</i> : $\mathcal{C}(\text{sense})$                                       |
| directed                             | bool           | <i>derived</i> : whether endpoint order is meaningful                              |
| ordering_only                        | bool           | <i>derived</i> : carries no truth coupling, produces no factor                     |
| exhaustive                           | bool           | <i>derived</i> : true iff the coupling is EXCLUSIVE                                |
| intended_strength_—band              | string         | strong, moderate, or weak                                                          |
| strength_range                       | [float, float] | the band’s numeric interval                                                        |
| validity                             | string         | valid or invalid                                                                   |
| error_kind                           | string null    | for invalid edges: wrong_sense, wrong_direction, false_endpoint, spurious          |
| is_concession                        | bool           | whether the relation is a conceded tension                                         |
| is_resolved_—concession              | bool           | whether the text resolves it                                                       |
| resolver_atom_id                     | string null    | the holding claim that resolves it                                                 |
| position_distance                    | int            | signed target-minus-source position gap                                            |
| window_admission                     | string         | whether the pair falls inside the locality window                                  |
| provenance                           | object         | which stage planned or transformed the edge                                        |
| <i>Negative set and expectations</i> |                |                                                                                    |
| non_relations                        | list           | declared-unrelated claim pairs with their position distance                        |
| expected.family_id,                  | string         | ladder membership and type                                                         |
| expected.family                      |                |                                                                                    |
| expected.rung_index,                 | int, string    | position in the ladder                                                             |
| expected.rung_name                   |                |                                                                                    |
| expected.perturbation                | object         | the calls applied, their targets, their parent rung, and the declared edge_effects |
| meta                                 | object         | canonical topic, domain, framing, split, word count, generator                     |

that dip positively rather than treating it as noise. A benchmark that only asserted monotonicity could not express this, and would score a correctly-behaving measure as failing.

### 9.3 METRICS OVER RELATION GRAPHS

Mined relations are matched to gold at three nested levels: **pair**, which asks only whether the two claims were related at all; **coupling**, which additionally requires the Level-1 coupling with direction required for the asymmetric couplings; and **sense**, which additionally requires the Level-2 sense.

Table 7: The per-rung relation-graph fix, holding the generator, the topics, the claim count and every prompt constant. Only the derivation of each rung’s gold graph changed (§5.6).

| Property                                             | Before       |                       | After        |
|------------------------------------------------------|--------------|-----------------------|--------------|
| Relations per item                                   | uniformly 10 | 8–11, varying by rung |              |
| Distinct edge sets per family (of 5 rungs)           | 1            |                       | 5            |
| Adjacent rung pairs with identical edge sets         | 4 of 4       |                       | 0 of 4       |
| Ordering constraints satisfied, gold arm, per family | 1/14         |                       | 11/14        |
| Ordering constraints satisfied, gold arm, corpus     | 2/28         |                       | <b>22/28</b> |

Coupling is the level to report, since it is the level at which a scorer selects a factor, and it is materially stricter than pair level.

Beyond micro-averaged precision, recall and F1 at each level, the harness stratifies recall by properties of the gold relation itself: by whether it runs forward or backward in claim order, by whether its coupling is directed, by its locality-window admission label, and by its validity. The stratification is not decoration. Forward/backward recall is what exposes a candidate-pair policy that cannot emit backward arcs at all, and validity-stratified recall must be read *inverted* on the invalid subset — those relations are planted errors, so a grounded extractor *failing* to recover them is arguably correct, and lower is better.

Precision against the declared non-relations is reported separately from precision against all unlabelled pairs, for the reason given in §5: the former has a real negative denominator and the latter does not. The harness also counts edges outside the nine admissible (sense, coupling) combinations, which cannot be scored against gold at all and are pure false positives.

Finally, the ordering-constraint checker reads each family’s manifest, evaluates every declared C1, C2 and C3 assertion against the scored rungs, and reports both the per-constraint outcome and the fraction satisfied. It records the observed direction and magnitude alongside the expectation, so a near-miss on a strict-increase constraint is distinguishable from an inversion.

## 10 VALIDATING THE BENCHMARK

A benchmark paper owes two demonstrations: that the instrument discriminates, and that its own construction is sound. We give the second first, because in our case it nearly failed.

**Scale and sampling, stated first.** All numbers below are on the released corpus: two CONFLICT families, ten items, sixteen claims each, with relations mined by llama-3.3-70b-instruct. Mining is sampled once per cell at the model’s default temperature, and repeated runs of an identical configuration moved coupling-level F1 by up to 0.06; across five independent samples the two revised extraction strategies span 0.49–0.58, every one above the unrevised 0.36, so the gap to the original exceeds the sampling noise while the ordering *between* the two revisions does not. With 28 ordering assertions per arm, the constraint-satisfaction figures are the only comparison here with enough observations to be more than indicative. These are measurements, not significance claims.

### 10.1 THE PER-RUNG RELATION-GRAPH DEFECT

An earlier version of the generator perturbed each rung’s *text* but re-attached the base plan’s relation graph to all five rungs, and passed every gate while doing so. The consequence is worth stating carefully because it is the failure mode a benchmark of this shape is most exposed to. If every rung of a family carries identical gold relations, then any measure computed *from gold relations* returns an identical score for all five rungs — so the strict-increase constraints of C1 cannot hold, not because the measure is wrong but because the corpus cannot express the difference it declares. The benchmark was silently untestable from gold, and the symptom was a gold-arm constraint satisfaction of 2/28: not a measure failing, but an instrument reading nothing.

Table 7 reports the fix: deriving each rung’s edges from its own operator calls (§5.6), assigning each call a distinct target, and adding the adjacency gate. Everything else is held constant — same

Table 8: Relation extraction against gold, over 86 edge-producing gold relations, micro-averaged. **Fwd/Bwd** split coupling-level recall by relation direction in claim order. **Non-rel** counts violations of the 48 declared non-relations. **Illegal** counts edges outside the nine admissible (sense, coupling) combinations. **Calls** is LLM calls issued.

| Extraction policy | Edges | Pair R | Coup P | Coup R | Coup F1     | Fwd  | Bwd  | Non-rel     | Calls |
|-------------------|-------|--------|--------|--------|-------------|------|------|-------------|-------|
| windowed          | 190   | 83.7   | 25.8   | 57.0   | 35.5        | 75.0 | 23.3 | 7/48        | 454   |
| gated             | 321   | 83.7   | 14.6   | 54.7   | 23.1        | 69.6 | 26.7 | 8/48        | 1062  |
| all pairs         | 867   | 95.3   | 9.3    | 82.6   | 16.8        | 87.5 | 73.3 | 10/48       | 3267  |
| bidirectional     | 205   | 83.7   | 39.0   | 66.3   | 49.1        | 71.4 | 56.7 | 5/48        | 505   |
| bidirectional v2  | 173   | 73.3   | 48.3   | 67.4   | <b>56.3</b> | 85.7 | 33.3 | <b>0/48</b> | 473   |

Table 9: Declared ordering constraints satisfied, 28 assertions per arm (14 per family  $\times$  two families). GOLD\_VALID is near-flat by construction — a conflict-fixing rung removes the planted-invalid conflicts this arm has already discarded — so its shortfall is expected and carries no information about the measure; GOLD is what the constraints test.

| Arm                     | f001  | f002  | % satisfied |
|-------------------------|-------|-------|-------------|
| GOLD                    | 11/14 | 11/14 | <b>78.6</b> |
| GOLD_VALID              | 7/14  | 4/14  | 39.3        |
| MINED, windowed         | 4/14  | 9/14  | 46.4        |
| MINED, gated            | 7/14  | 8/14  | 53.6        |
| MINED, all pairs        | 7/14  | 5/14  | 42.9        |
| MINED, bidirectional    | 5/14  | 5/14  | 35.7        |
| MINED, bidirectional v2 | 6/14  | 8/14  | 50.0        |

generator, same two canonical topics, same sixteen claims per item, five distinct response texts per family in both versions. Gold-arm agreement moves from 2/28 to 22/28.

We report this at length for three reasons. It is the strongest available evidence that the ladder is a genuine test of a measure rather than a measurement of the corpus: after the fix each rung’s labels describe the response that ships with them, so the constraints have something to be right or wrong about. It establishes the diagnostic value of the gold arm — an instrument whose gold arm scores near zero is broken, whatever its mined arms report. And it is a warning to anyone building a benchmark of this shape: the defect passed every content gate, because every individual item was well-formed and only the *relationship between* items was degenerate.

## 10.2 THE BENCHMARK DISCRIMINATES BETWEEN RELATION EXTRACTORS

Table 8 shows the corpus separating five extraction configurations across a range of 39.5 F1 points, which is the primary evidence that the annotation is informative. Three findings are corpus properties rather than facts about any one extractor.

The declared non-relations work as intended. Violations range from 10/48 to 0/48 across configurations, and the best-precision configuration is the only one that reaches zero — so this small, explicitly-labelled negative set ranks the configurations in the same order as coupling precision while being interpretable in a way that precision against unlabelled pairs is not.

The locality metadata exposes a structural limitation rather than a tuning one. Backward-running gold relations are recovered at 23–27% by the three forward-only policies and 73.3% by all-pairs, because 20 of the 86 gold relations are both directed and backward and a forward-only candidate set cannot emit them at all. A benchmark that did not record each relation’s signed position distance could not distinguish this from a model failure.

The admissibility metadata catches a failure mode that F1 alone hides. Illegal-vocabulary emissions — edges whose (sense, coupling) pair no reading of the scheme licenses — run from 93 down to 0 across configurations. These cannot be scored against gold at all, so an extractor emitting many of them is being flattered by any metric computed over the scorable remainder.

Table 9 gives the ordering results. The gold arm satisfies 78.6% of declared constraints and the mined arms 35.7–53.6%. Since the arms differ in nothing but the relation graph, the 25–43 point gap is the extraction contribution, isolated by construction — and it is consistent with Table 8, since at 56.3% coupling F1 roughly half the relations in a mined graph are wrong or missing, and a ladder whose consecutive rungs differ by one or two relations cannot survive that.

We are explicit about what these numbers do *not* establish. The ordering of the mined arms does not track their coupling F1 — the gated policy scores best among them at the second-worst F1 — and at 28 assertions per arm we read that as sampling noise rather than signal. Ranking extraction policies on ladder agreement requires a substantially larger corpus, and Table 8 is the better instrument for that comparison today.

**Why the gold arm is not at 100%.** Six of the 28 gold-arm constraints fail, and each is attributable to a specific declared assertion rather than to a diffuse inadequacy. Three are C1 strict-increase assertions on the mean-marginal readout where the observed change is  $-0.000$  to three decimal places: an *invariance* where a strict increase was declared. Those three are statements about the declared expectation — the perturbation at that rung pair happens not to move that particular readout — rather than inversions. Two are C2 assertions where a predicted invariance of the log-partition readout came out as a decrease of  $-0.000$  and  $-0.002$ , again at the resolution limit. The sixth is a genuine inversion: a C3 endpoint separation on the consistency readout,  $-0.004$  in the wrong direction, which is exactly the failure mode §9 anticipates for that readout on families whose conflicts are all EXCLUSIVE. A benchmark whose gold arm scored 28/28 would have constraints too weak to be informative; the useful property is that five of the six failures are resolution-limited near-misses and the sixth is a predicted defect of one readout.

## 11 LIMITATIONS

**Scale.** The released corpus is two families and ten items against a design target of 120 families. Every figure in §10 should be read at that scale: the 28 ordering assertions per arm are enough to establish that the instrument discriminates and not enough to rank extraction policies, and single-sample mining with measured run-to-run movement of 0.06 F1 means the second decimal of any single cell is not meaningful.

**Three of four ladder types are ungenerated.** CHAIN, ORDER and CONTROL are specified, implemented and gate-checked, but not generated at scale, so no result here speaks to them. The most consequential absence is CONTROL, whose entire purpose is to catch a measure that responds to having been edited rather than to coherence — arguably the single most important experiment the protocol specifies. Until it is run, a measure can satisfy every ordering constraint we report while being a sophisticated edit-counter, and this benchmark would not know. Similarly, CHAIN is the type on which contradiction-detection approaches should fail, so its absence means we have not yet demonstrated the corpus separates the two mechanisms it is designed to separate.

**The corpus is LLM-generated, including its labels.** Questions, claims, relation plans, responses, perturbations and gold relation labels are all model output, validated by a model committee. The structural mitigations of §6 address the most obvious circularity — the generator is excluded from its own committee, and V1 requires an independent model to recover the planned relations from the prose before admission — but they do not address a bias *shared* between generator and committee, and current frontier models are not independent in the relevant sense. Human verification of a sample is required before the corpus is treated as ground truth, and we have not performed it. This is the limitation we would most want addressed before the benchmark is used to make comparative claims about published measures.

**Declared orderings are assumptions, not observations.** A ladder’s constraints encode our judgement that, for instance, resolving a conceded tension increases coherence. That judgement is defensible and is the premise of the whole construction, but it is a premise: a measure could satisfy every constraint and still disagree with human readers if our declared orderings are not the ones readers perceive. The remedy is human coherence ratings on the same items, correlated against the measures — which the corpus supports and which we have not collected.



**Genre and length are narrow.** Every item is 550–650 words of formal expository prose answering an analytical question, with 14–16 claims. Conclusions do not transfer automatically to dialogue, to summarization of a source document, to much longer arguments where a conflict may span thousands of words, or to informal registers where relations are signalled differently. The locality bias of P3 — preferring related pairs within four positions — additionally means the corpus under-represents genuinely long-range discourse dependencies.

**One generator, one extraction model.** The released corpus comes from a single generating model and its mined arms from a single extraction model. Generator-specific artifacts are therefore not controlled for by variation, only by the V3 leakage gate, and the extraction results characterize one model rather than the state of the art.

## 12 CONCLUSION

Evaluating a coherence measure requires contrasts that vary coherence while holding content fixed, and ordinary corpora do not provide them. LoCoBench provides them as ladders: families of responses over one claim set, with a declared relative ordering, per-rung machine-readable relation graphs, an explicitly labelled negative set, and a family type whose required pattern is no trend at all. The construction makes two things possible that a rating-correlation study cannot. It attributes a score difference to coherence rather than to content, because the content is held constant. And by scoring a measure twice — once on its own extracted relations and once on gold — it isolates extraction error from scoring error by construction rather than estimating it after the fact.

Applying that separation to the released corpus locates the field’s bottleneck. Measures scored from gold relation graphs satisfy 78.6% of declared ordering constraints; the same measures scored from graphs mined out of the response text satisfy 35.7–53.6%, and the best extraction configuration reaches 56.3% coupling-level F1. The scoring models are adequate to the task as posed, and grounded relation extraction is where the loss is. That is a narrower and more tractable target than “measure coherence”.

We also report a construction defect at length, because it generalizes. An earlier version of the corpus attached the same relation graph to every rung of each family, passed every content gate while doing so, and was thereby rendered untestable from gold — 2/28 constraints satisfied, rising to 22/28 once each rung’s graph was derived from its own perturbations. Every individual item was well-formed; only the relationship between items was degenerate. Any benchmark that grades a measure on comparisons rather than on absolute values is exposed to the same class of defect, and the gold arm is the instrument that detects it.

### AI USE STATEMENT

This is a benchmark constructed *by* generative AI, and the disclosure is correspondingly extensive. Every artifact in the released corpus — the questions, the atomic claims, the relation plans, the response prose, the perturbed rungs, and the gold relation labels — is the output of a large language model driven by the prompts of Appendix A, and every acceptance decision is the majority vote of a committee of three further language models. §11 states plainly what this does and does not establish, and we regard human verification of a sample as a precondition for treating the corpus as ground truth. Separately, we used AI assistants for code implementation and for editorial assistance on this manuscript. All LLM-generated code was reviewed by the authors and is covered by a test suite that asserts, among other things, the instruction count and placeholder set of each of the eight prompts against the specification this paper reports. The benchmark design — the ladder construction, the constraint classes, the operator inventory, the gate thresholds and their calibration, the schema and the metrics — is the authors’ own. We take responsibility for the final content of this work.

### ETHICS STATEMENT

**Intended use.** LoCoBench is intended for evaluating measures of logical coherence in long-form generated text. It is not a factuality benchmark and must not be used as one: coherence and factuality are distinct properties, and a response can be perfectly coherent and entirely false. It is not a training corpus for text generation, and at ten items it is not suitable as one.

**Deliberately false content.** Every item asserts at least two factually false claims, and every relation plan contains four deliberately invalid relations. This is essential to the design — without false content, coherence and factuality would be perfectly correlated in the data — and it makes the corpus hazardous if mishandled. The false claims are labelled at the claim level and the invalid relations at the edge level, and **the corpus must not be redistributed with those labels stripped**. Extracted prose from these items should not be treated as a source of factual information on any topic.

**Human subjects and content.** No human subjects were involved and no personally identifying data is present. Topics span academic and professional domains; the adjudicative framing of many items concerns generic incident and dispute scenarios rather than real cases or identifiable individuals.

**Compute.** Generating one family requires on the order of a few dozen model calls across generation and validation, and a scored mined arm on ten items requires several hundred more. This is modest, but it is not free, and a benchmark that requires an LLM committee to extend is less accessible than one that does not — a real cost of the validation design we chose.

#### REPRODUCIBILITY STATEMENT

The pipeline is specified in the paper and the prompts are reproduced verbatim. Appendix A gives all eight, extracted programmatically from the released source so that the appendix cannot drift from the code, each with its character count, instruction count and placeholder set; Appendix B gives the specification table these are asserted against and the substitution contract. Table 4 gives every acceptance threshold in one place, Table 3 the perturbation operators, Table 6 the item schema field by field, and §5.6 the derivation of each rung’s gold graph. §7 names the generating and committee models and the topics used. The release comprises the items, the family manifests with their declared constraints, the generation log, and the rejected families with their rejection reasons, so that reported admission rates are auditable. The gold-arm evaluation requires no LLM access and is deterministic.

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## A THE GENERATION AND VALIDATION PROMPTS, VERBATIM

All eight prompts follow, reproduced exactly as issued and extracted programmatically from the released source. Placeholders have the form [NAME\_PLACEHOLDER] and are substituted by the contract of Appendix B. Line numbers are added for reference and are not part of the prompt.

### A.1 P1 — TOPIC TO QUESTION

1,904 characters, 38 lines, 9 numbered instructions. Placeholder: [TOPIC\_PLACEHOLDER].

```

1 Instructions:
2 1. You are given a TOPIC. Generate one open-ended QUESTION about that topic.
3 2. The QUESTION must begin with Why, How, What, Does, or Which.
4 3. The QUESTION must elicit an answer containing MULTIPLE, DIVERSE logical
5 relationships. Aim for an answer that would naturally contain at least four
6 of the following: a causal chain; a condition; a temporal ordering; an
7 evidential appeal; a general claim with a specific instance; a restatement
8 or definitional equivalence; a non-exhaustive contrast; an EXHAUSTIVE pair
9 of competing alternatives of which EXACTLY ONE can hold; a disjunctive pair
10 of which AT LEAST ONE must hold; a tension that an authoritative finding
11 later resolves.
12 4. Prefer topics where competing explanations are adjudicated: incident
13 investigations, litigation, diagnostic reasoning, attribution disputes,
14 competing scientific hypotheses. Such topics naturally produce exhaustive
15 alternatives and resolving holdings.
16 5. The QUESTION must be answerable in 500-900 words of formal prose.
17 6. The QUESTION must not itself contain the words "cause", "condition",
18 "alternative", or "contrast", so the answer's relations are not primed
19 lexically.
20 7. Do NOT ask a multi-part question joined by "and" - one question only.
21 8. Wrap the QUESTION in square brackets.
22 9. Output the bracketed QUESTION only, with no preamble.
23
24 Use the following examples to better understand your task.
25
26 TOPIC: aviation component failure investigation
27 QUESTION: [How do investigators determine responsibility when a component
28 supplied by a third party fails in service and the operator and the
29 manufacturer offer competing explanations?]
30
31 TOPIC: antibiotic resistance in hospital settings
32 QUESTION: [Why does resistance emerge in some hospital units but not others,
33 and how is a specific outbreak traced to its origin?]
34
35 Your Task:
36 TOPIC: [TOPIC_PLACEHOLDER]
37 QUESTION:
38
```

### A.2 P2 — QUESTION TO TAGGED CLAIMS

3,754 characters, 66 lines, 7 numbered instructions. Placeholder: [QUESTION\_PLACEHOLDER].

```

1 Instructions:
2 1. You are given a QUESTION. Generate a set of ATOMIC CLAIMS that a
3 well-informed answer to it might contain.
4 2. An atomic claim is a sentence containing a singular piece of information,
5 which should be basic, indivisible, and independently verifiable.
6 3. Generate exactly 26 atomic claims: the 24 below, plus the 2 holding claims
7 required by instruction 4. The 24 break down as:
8 (a) 14 factually correct claims;
9 (b) 4 factually incorrect claims (plausible but false);
10 (c) 2 EXHAUSTIVE ALTERNATIVE claims: a pair X, Y such that exactly one of X
11 and Y must be true - they cannot both hold, and they cannot both fail.
12 Example pair: "No one was harmed in the incident." / "Three people died
13 in the incident."
14 (d) 2 DISJUNCTIVE claims: a pair X, Y such that at least one must be true,
15
```

1026 16 but both may be true simultaneously. Make each one a STATE OF THE WORLD,  
 1027 17 not a finding or a check that reported something: "the alloy was  
 1028 18 substituted" and "the batch was mislabelled", not "assay A flagged it"  
 1029 19 and "assay B flagged it".  
 1030 20 (e) 2 EQUIVALENT claims: a pair X, Y asserting the same fact in different  
 1031 21 words, so that either one being true makes the other true.  
 1032 22 4. Additionally generate 2 HOLDING claims: sentences in which an authority  
 1033 23 (a court, tribunal, regulator, final report, or review board) reaches a  
 1034 24 determination that settles a dispute between two other claims in your list.  
 1035 25 A holding claim must use adjudicating language - "held", "ruled",  
 1036 26 "concluded", "determined", "found that", "ultimately".  
 1037 27 5. Each claim must be a single declarative sentence with no connective linking  
 1038 28 it to another claim. Do NOT use "because", "although", "therefore", "if",  
 1039 29 "while", "however". Relations are added at a later stage, not here.  
 1040 30 6. Do NOT indicate which claims are factually incorrect in their wording; do  
 1041 31 NOT hedge with "assume", "might", "possibly", or "allegedly".  
 1042 32 7. Output as a markdown code block: an unordered list with "-" bullets. Append  
 1043 33 a tag in square brackets to every claim, one of: [correct], [incorrect],  
 1044 34 [alt-pair-1], [alt-pair-2], [disj-pair-1], [disj-pair-2], [equiv-pair-1],  
 1045 35 [equiv-pair-2], [holding].  
 1046 36  
 1047 37 Use the following example to better understand your task.  
 1048 38  
 1049 39 QUESTION: [How do investigators determine responsibility when a component  
 1050 40 supplied by a third party fails in service and the operator and the  
 1051 41 manufacturer offer competing explanations?]  
 1052 42  
 1053 43 ATOMIC CLAIMS:  
 1054 44 ...  
 1055 45 - AeroParts supplied turbine blades to the carrier. [correct]  
 1056 46 - The turbine blades were installed in the carrier's A-320 fleet. [correct]  
 1057 47 - Seventeen turbine blades failed in flight during service. [correct]  
 1058 48 - The regulator opened a safety probe into the blade failures. [correct]  
 1059 49 - Maintenance logs recorded abnormal vibration on the failed blades. [correct]  
 1060 50 - The airline argued that the failures were caused by pilot error. [correct]  
 1061 51 - The final report concluded the root cause was a metallurgical defect. [correct]  
 1062 52 - The regulator issued an order grounding the fleet. [correct]  
 1063 53 - Every affected aircraft was pulled from service. [equiv-pair-1]  
 1064 54 - The regulator's order removed all affected aircraft from operation. [equiv-pair-2]  
 1065 55 - The blades were certified under a 1998 airworthiness standard. [incorrect]  
 1066 56 - No one was harmed in the incidents. [alt-pair-1]  
 1067 57 - Three passengers died in one of the failures. [alt-pair-2]  
 1068 58 - The alloy supplied for the blades was substituted. [disj-pair-1]  
 1069 59 - The blade batch was mislabelled at the mill. [disj-pair-2]  
 1070 60 - The tribunal held that AeroParts was liable for the blade failures. [holding]  
 1071 61 - The accident report found the pilots not at fault. [holding]  
 1072 62 ...  
 1073 63  
 1074 64 Your Task:  
 1075 65 QUESTION: [QUESTION\_PLACEHOLDER]  
 1076 66 ATOMIC CLAIMS:

### A.3 P3 — CLAIMS TO RELATION PLAN

7,495 characters, 130 lines, 10 numbered instructions. Placeholders:  
 [QUESTION\_PLACEHOLDER], [CLAIMS\_PLACEHOLDER]. The longest prompt in the  
 pipeline, and the one carrying the plan's structural quotas.

1 Instructions:  
 2  
 3 1. You are given a QUESTION and a tagged list of ATOMIC CLAIMS. Produce a  
 4 RELATION PLAN: a machine-readable graph of the logical relations a response  
 5 should draw between those claims.  
 6  
 7 2. Select 14-16 claims to use. Aim for 15. Number them with FRESH consecutive  
 8 positions: the first claim you assert is pos 1, the next pos 2, and so on  
 9 with NO gaps and NO repeats, ending at pos N where N is how many you  
 10 selected. Do NOT carry over the numbering of the input list - if you pick the  
 11 3rd, 7th and 12th claims they become pos 1, 2 and 3. The position is the  
 12 order the response will assert the claim in. Every selected claim keeps its  
 13 text EXACTLY as given. Do NOT introduce information not in the claims, and  
 14 do NOT alter the claims. Your selection MUST INCLUDE AT LEAST 2 of the  
 15 [incorrect] claims: the response has to assert false content alongside the  
 16 true, so a selection drawn only from the [correct] claims is rejected. Carry  
 17 each selected claim's tag through as a "factual" field - false for an  
 18 [incorrect] claim, true for every other tag.  
 19 3. Plan EXACTLY 10 RELATIONS. Each relation connects exactly two selected claims  
 and has a SENSE. Every relation must join a DIFFERENT pair of claims: once two

claims are related, that pair is used up and must not appear again in the relations list, in either direction. The SENSE is one of: Cause-Effect, Effect-Cause, Evidence, Condition, Restatement, Instantiation, Contrast, Concession, Alternative, Disjunction, Precedence, Succession.

- Cause-Effect: source causes target. Effect-Cause: source is the effect, target its cause.
- Evidence: source is evidence for target. Condition: source is a condition for target.
- Restatement: source and target assert the same thing.
- Instantiation: source is a general claim, target a specific instance.
- Contrast: opposition that is NOT exhaustive - both could be false.
- Concession: a tension the text itself resolves.
- Alternative: EXHAUSTIVE competing options - EXACTLY ONE holds. Not both, and not neither. Use your [alt-pair-\*] claims here.
- Disjunction: AT LEAST ONE holds; both may hold. Use your [disj-pair-\*] claims here.
- Precedence / Succession: ordered in time with NO truth dependence - neither claim being true or false says anything about the other.

4. Keep relations LOCAL: prefer pairs within 4 POSITIONS of each other, so the graph stays recoverable from the response by a local reader. A longer-range pair is acceptable when the two claim texts share a named entity, and a few long-range links (for instance a closing claim answering the opening one) are fine. Order the positions so that related claims sit near each other.
5. The plan must include AT LEAST ONE relation of each of these senses: Alternative, Disjunction, Restatement, Concession, and one of Precedence/Succession.
6. Every Concession must be marked resolved or unresolved. A resolved Concession must name a RESOLVER: a [holding] claim, also selected and positioned AFTER both endpoints.
7. Give each relation an intended STRENGTH BAND: strong (0.85-1.00), moderate (0.60-0.84), or weak (0.35-0.59) - how strongly the source determines the target under that sense. Restatement relations are always strong.
8. Give each relation an intended VALIDITY: valid if the relation genuinely holds between the claims as stated; invalid if it does not. Of your 10 relations, mark EXACTLY 6 "valid" and EXACTLY 4 "invalid" - count them before you emit, and do not compute a percentage. For each invalid relation give an ERROR\_KIND, one of: wrong\_sense, wrong\_direction, false\_endpoint, spurious.
9. Also list 4-6 NON-RELATIONS (aim for 5): pairs of selected claims within 4 positions of each other between which the response must draw NO logical dependence. These are the distractors a relation miner must correctly label None. A non-relation pair must NOT also appear in your relations list, in either direction - the two lists are disjoint.
10. Output a single JSON object in a code block and nothing else. Below is a COMPLETE, correctly-shaped example - copy its structure exactly, not its text. It has 14 atoms numbered 1..14 with no gaps, 2 of them factually incorrect, 10 relations (6 valid, 4 invalid), all five required senses, and 5 non-relations disjoint from the relations:

```

'''json
{
  "atoms": [
    {"pos": 1, "text": "<claim text>", "factual": true},
    {"pos": 2, "text": "<claim text>", "factual": true},
    {"pos": 3, "text": "<claim text>", "factual": true},
    {"pos": 4, "text": "<claim text>", "factual": false},
    {"pos": 5, "text": "<claim text>", "factual": true},
    {"pos": 6, "text": "<claim text>", "factual": true},
    {"pos": 7, "text": "<claim text>", "factual": true},
    {"pos": 8, "text": "<claim text>", "factual": true},
    {"pos": 9, "text": "<claim text>", "factual": false},
    {"pos": 10, "text": "<claim text>", "factual": true},
    {"pos": 11, "text": "<claim text>", "factual": true},
    {"pos": 12, "text": "<claim text>", "factual": true},
    {"pos": 13, "text": "<claim text>", "factual": true},
    {"pos": 14, "text": "<claim text>", "factual": true}
  ],
  "relations": [
    {"source_pos": 1, "target_pos": 2, "sense": "Cause-Effect",
     "strength_band": "strong", "validity": "valid",
     "error_kind": null, "resolved": null, "resolver_pos": null},
    {"source_pos": 2, "target_pos": 3, "sense": "Evidence",
     "strength_band": "moderate", "validity": "valid",
     "error_kind": null, "resolved": null, "resolver_pos": null},
    {"source_pos": 3, "target_pos": 4, "sense": "Restatement",
     "strength_band": "strong", "validity": "valid",
     "error_kind": null, "resolved": null, "resolver_pos": null},
    {"source_pos": 5, "target_pos": 6, "sense": "Alternative",
     "strength_band": "strong", "validity": "valid",
     "error_kind": null, "resolved": null, "resolver_pos": null},
    {"source_pos": 7, "target_pos": 8, "sense": "Disjunction",
     "strength_band": "strong", "validity": "valid",
     "error_kind": null, "resolved": null, "resolver_pos": null},
    {"source_pos": 9, "target_pos": 10, "sense": "Contrast",
     "strength_band": "moderate", "validity": "invalid",
     "error_kind": "wrong_sense", "resolved": null, "resolver_pos": null},
    {"source_pos": 11, "target_pos": 12, "sense": "Concession",
     "strength_band": "weak", "validity": "invalid",
     "error_kind": "wrong_sense", "resolved": null, "resolver_pos": null},
    {"source_pos": 13, "target_pos": 14, "sense": "Precedence",
     "strength_band": "weak", "validity": "invalid",
     "error_kind": "wrong_sense", "resolved": null, "resolver_pos": null},
    {"source_pos": 1, "target_pos": 14, "sense": "Disjunction",
     "strength_band": "strong", "validity": "invalid",
     "error_kind": "wrong_direction", "resolved": null, "resolver_pos": null},
    {"source_pos": 2, "target_pos": 13, "sense": "Disjunction",
     "strength_band": "strong", "validity": "invalid",
     "error_kind": "wrong_direction", "resolved": null, "resolver_pos": null}
  ],
  "non_relations": [
    {"source_pos": 1, "target_pos": 1, "sense": null, "strength_band": null, "validity": null, "error_kind": null, "resolved": null, "resolver_pos": null},
    {"source_pos": 1, "target_pos": 14, "sense": null, "strength_band": null, "validity": null, "error_kind": null, "resolved": null, "resolver_pos": null},
    {"source_pos": 14, "target_pos": 1, "sense": null, "strength_band": null, "validity": null, "error_kind": null, "resolved": null, "resolver_pos": null},
    {"source_pos": 14, "target_pos": 14, "sense": null, "strength_band": null, "validity": null, "error_kind": null, "resolved": null, "resolver_pos": null},
    {"source_pos": 1, "target_pos": 13, "sense": null, "strength_band": null, "validity": null, "error_kind": null, "resolved": null, "resolver_pos": null}
  ]
}

```

```

1134 101     "strength_band": "moderate", "validity": "valid",
1135 102     "error_kind": null, "resolved": null, "resolver_pos": null},
1136 103     {"source_pos": 9, "target_pos": 10, "sense": "Concession",
1137 104     "strength_band": "moderate", "validity": "valid",
1138 105     "error_kind": null, "resolved": true, "resolver_pos": 11},
1139 106     {"source_pos": 11, "target_pos": 12, "sense": "Precedence",
1140 107     "strength_band": "weak", "validity": "invalid",
1141 108     "error_kind": "wrong_sense", "resolved": null, "resolver_pos": null},
1142 109     {"source_pos": 12, "target_pos": 13, "sense": "Cause-Effect",
1143 110     "strength_band": "moderate", "validity": "invalid",
1144 111     "error_kind": "wrong_direction", "resolved": null, "resolver_pos": null},
1145 112     {"source_pos": 13, "target_pos": 14, "sense": "Contrast",
1146 113     "strength_band": "weak", "validity": "invalid",
1147 114     "error_kind": "spurious", "resolved": null, "resolver_pos": null},
1148 115     {"source_pos": 4, "target_pos": 6, "sense": "Instantiation",
1149 116     "strength_band": "weak", "validity": "invalid",
1150 117     "error_kind": "false_endpoint", "resolved": null, "resolver_pos": null}
1151 118   ],
1152 119   "non_relations": [
1153 120     {"source_pos": 1, "target_pos": 4}, {"source_pos": 2, "target_pos": 5},
1154 121     {"source_pos": 6, "target_pos": 9}, {"source_pos": 8, "target_pos": 11},
1155 122     {"source_pos": 10, "target_pos": 13}
1156 123   ]
1157 124 }
1158 125 ...
1159 126
1160 127 Your Task:
1161 128 QUESTION: [QUESTION_PLACEHOLDER]
1162 129 ATOMIC CLAIMS: [CLAIMS_PLACEHOLDER]
1163 130 RELATION PLAN:

```

#### A.4 P4 — PLAN TO RESPONSE

5,573 characters, 84 lines, 8 numbered instructions. Placeholders: [QUESTION\_PLACEHOLDER], [PLAN\_PLACEHOLDER].

```

1164 1
1165 2 Instructions:
1166 3 1. You are given a QUESTION and a RELATION PLAN in JSON. Write a RESPONSE that
1167 4   answers the question and realizes the plan exactly.
1168 5 2. Assert every planned atom EXACTLY ONCE, in the plan's POSITION order,
1169 6   preserving each atom's factual content. Do not restate an atom you have
1170 7   already asserted in order to attach a later relation to it - refer back with
1171 8   a pronoun or a short noun phrase instead. Name that phrase after the thing
1172 9   the atom is about, not after its status as a finding: "the primitive carpal
1173 10  morphology" or "that wrist evidence", not "the analysis's flagging of
1174 11  primitive traits"; "these two accounts" or "either possibility", not "the two
1175 12  claims in circulation". A summary phrase covering both endpoints of one
1176 13  connective is fine - "at least one of these must hold: X, or Y". You may
1177 14  adjust surface wording for fluency but must not change what is asserted.
1178 15  Quantifiers, determiners, numbers, dates, polarity and modals are CONTENT,
1179 16  not wording: keep "all", "only", "no", "some", "each", "never" and every
1180 17  figure exactly as the atom has them. "All tools were crafted by a single
1181 18  culture" may become "every one of the tools came from a single culture", but
1182 19  never "the tools were crafted by a single culture" - dropping "All" weakens
1183 20  the claim and counts as changing it.
1184 21  Never let one atom absorb another: each must stay separately true or false.
1185 22  Joining two atoms with a connective is fine and often required - "either X or
1186 23  Y" and "although X, Y" both assert two atoms and are exactly what instruction
1187 24  3 asks for.
1188 25 3. Realize every planned RELATION so a careful reader could recover it from the
1189 26  text alone, using language appropriate to its sense:
1190 27   - Cause-Effect / Effect-Cause: "caused", "led to", "resulted from", "because"
1191 28   - Evidence: "indicates", "confirms", "as shown by"
1192 29   - Condition: "if", "provided that", "only when"
1193 30   - Restatement: "that is", "in other words", "equivalently"
1194 31   - Instantiation: "for example", "specifically", "in one case"
1195 32   - Contrast: "by contrast", "whereas", "on the other hand"
1196 33   - Concession: "although", "despite", "even though"
1197 34   - Alternative: make the exhaustiveness EXPLICIT - "either ... or", "one of
1198 35     these accounts must be wrong", "the two cannot both be true, and one of
1199 36     them must hold"
1200 37   - Disjunction: make the at-least-one reading EXPLICIT - "at least one of",
1201 38     "one or both", "perhaps both". Say "perhaps both", never "possibly both".
1202 39   A bare "X, or Y" is NOT enough for either of these two senses: on its own it
1203 40   leaves a reader unable to tell whether both claims could hold, which is the
1204 41   only thing separating Alternative from Disjunction. Every such pair needs its
1205 42   marker - "either X or Y" when exactly one holds, "at least one of X or Y,

```

perhaps both" when both may.

- Precedence / Succession: state the sequence EXPLICITLY in one sentence joining both claims - "before", "after", "subsequently", "earlier than", "which postdates" - while keeping the ordering purely chronological, so a reader sees only when things happened and infers nothing from it. Write "X, which predates Y" or "Y came after X", not "X and therefore Y". Asserting the two claims in separate sentences leaves this relation unrealized: an ordering a reader cannot see is the same as no ordering.

4. For each planned NON-RELATION, assert both atoms but draw no dependence between them. Separate them by DISTANCE and PARAGRAPHING, not by a marker: put other material between them, or place them in different paragraphs, and simply use no connective. Do NOT announce the absence of a link - phrases like "In an unrelated matter", "Separately", or "On a different note" read as stitched-together annotation rather than prose and will be marked down.
5. For a resolved Concession, assert the resolver atom after both endpoints and present it as settling the tension.
6. Write it as a continuous expository answer to the QUESTION - an argument a domain expert would publish, organized into 3-5 paragraphs, in which the planned relations are simply how the argument hangs together. Do NOT walk the plan relation by relation: a reader must not be able to tell that a plan exists. Restate a subject to carry an argument forward ("the radiocarbon date", "that attribution"), never the earlier sentence as a step ("the finding just reported", "the provision just stated", "the ordering just described"). Use precise and formal language, avoiding vague generalizations and rhetorical fillers. Aim for 550-650 words. The 500-word floor is a hard requirement checked by an automated reader, which rejects a shorter answer outright however good it is, so give each atom and each relation the space to be argued rather than compressing them into a summary.
7. Do NOT mention the relation plan, the senses, the strength bands, the validity labels, or the fact that some content is incorrect. Do NOT hedge with "assume", "might", "possibly", "allegedly", "supposedly", "reportedly", or "it is claimed" around planned-invalid relations - assert them in the same register as the valid ones. Do NOT add headings, bullet lists, or annotations.
8. Wrap the RESPONSE in a code block whose contents are the prose itself, as you would publish it. The block holds the answer, not a JSON object or any other payload with the prose inside a field.

Your Task:  
 QUESTION: [QUESTION\_PLACEHOLDER]  
 RELATION PLAN: [PLAN\_PLACEHOLDER]  
 RESPONSE:

## A.5 P5 — RESPONSE TO PERTURBED RUNG

2,594 characters, 44 lines, 5 numbered instructions. Placeholders: [RESPONSE\_PLACEHOLDER], [PLAN\_PLACEHOLDER], [OPERATOR\_PLACEHOLDER]. Instruction 3 defines all ten perturbation calls of Table 3.

- 1 Instructions:
- 2
- 3 1. You are given a RESPONSE, its RELATION PLAN, and one PERTURBATION to apply.
- 4 2. Apply ONLY the named perturbation. Change the minimum number of words
- 5 needed. Every other relation, atom, and sentence must be preserved verbatim.
- 6 3. The perturbations are defined as follows.
- 7 - wrong\_sense(edge, new\_sense): rewrite the connective realizing that edge so
- 8 the text draws new\_sense instead. Keep both atoms' content unchanged.
- 9 - direction\_reversal(edge): rewrite so the relation runs from the target to
- 10 the source. Keep both atoms' content unchanged.
- 11 - exhaustiveness\_flip(edge): if the edge is Alternative, rewrite so the two
- 12 claims merely oppose without being exhaustive (both could be false). If it
- 13 is Contrast, rewrite so exactly one must hold.
- 14 - remove\_resolution(edge): delete the resolver sentence and any reference to
- 15 it, leaving the conceded tension unresolved. Change nothing else.
- 16 - inject\_contradiction(atom): add ONE new sentence asserting a claim that
- 17 directly contradicts the named atom. Place it at least two sentences later.
- 18 - break\_chain(edge): remove or negate the link realizing that edge so the
- 19 causal chain no longer connects, keeping both endpoint atoms present.
- 20 - shuffle\_order(level): reorder the response's sentences without changing any
- 21 wording. level=full permutes all sentences; level=block permutes paragraph
- 22 blocks; level=adjacent swaps one adjacent pair.
- 23 - spurious\_relation(atom\_a, atom\_b): add a connective asserting a causal or
- 24 contradictory link between two atoms the original response kept separate.
- 25 Add no new factual content.
- 26 - drop\_relation(edge): remove the connective realizing that edge so the two
- 27 atoms are merely adjacent assertions. Keep both atoms.
- 28 - ordering\_only(edge): rewrite a Precedence edge as Succession, or the

```

1242 29         reverse, preserving the actual chronology and adding NO causal or
1243 30         inferential language. The truth of neither atom may become dependent on
1244 31         the other.
1245 32 4. Do not change the response's length by more than 15 percent, except for
1246 33 inject_contradiction (which adds one sentence) and remove_resolution /
1247 34 break_chain (which remove one).
1248 35 5. Output the perturbed RESPONSE in a code block, then a second code block
1249 36 containing a JSON diff:
1250 37 {"operator": "...", "target": ..., "sentences_changed": [...],
1251 38  "relations_added": [...], "relations_removed": [...],
1252 39  "relations_relabeled": [...]}
1253 40
1254 41 Your Task:
1255 42 RESPONSE: [RESPONSE_PLACEHOLDER]
1256 43 RELATION PLAN: [PLAN_PLACEHOLDER]
1257 44 PERTURBATION: [OPERATOR_PLACEHOLDER]

```

## A.6 V1 — RELATION RECOVERY

4,088 characters, 67 lines. Placeholders: [RESPONSE\_PLACEHOLDER], [ATOMS\_PLACEHOLDER]. The three validators use a function-signature idiom rather than numbered instructions, so their instruction count is zero by construction and their contract is the signature line. Note that this prompt states the compile map explicitly in its final rule, so the coupling a validator emits is required to be the one its chosen sense compiles to.

```

1262 1
1263 2 Let's define a function named recover_relations(response: str,
1264 3 atoms: dict[str, str]).
1265 4 Given a RESPONSE and the ATOMS drawn from it, the function returns the
1266 5 list of logical relations the response ITSELF draws between those atoms.
1267 6
1268 7 ATOMS maps an atom's NUMBER to its text, numbered from 1. Use those numbers, as
1269 8 integers, to identify atoms in your output: if the relation holds between the
1270 9 atoms numbered "3" and "7", emit source 3 and target 7. Never renumber them and
1271 10 never count from zero.
1272 11
1273 12 For each ORDERED pair (i, j) that the response actually relates, emit one object:
1274 13 {"source": i, "target": j,
1275 14  "sense": one of ["Cause-Effect", "Effect-Cause", "Evidence", "Condition",
1276 15  "Restatement", "Instantiation", "Contrast", "Concession", "Alternative",
1277 16  "Disjunction", "Precedence", "Succession"],
1278 17  "coupling": one of ["entailment", "contradiction", "equivalence", "exclusive",
1279 18  "co_necessity", "none"],
1280 19  "strength_band": one of ["strong", "moderate", "weak"],
1281 20  "resolved": true or false or null}
1282 21
1283 22 Rules:
1284 23 - Emit a relation ONLY if the response itself draws that connection, as written
1285 24 or as a clear step in the author's argument. Do not emit a relation that is
1286 25 merely plausible in general.
1287 26 - Pick the MOST SPECIFIC sense the text supports, never the most general one.
1288 27 Five senses compile to "entailment", and "Evidence" is the broadest of them,
1289 28 so it is the easy wrong answer: use it only for evidential support ("indicates",
1290 29 "confirms", "as shown by", "suggests"). Prefer "Cause-Effect" when the text says
1291 30 one thing brought the other about ("caused", "led to", "resulted in",
1292 31 "because"), "Effect-Cause" when that runs the other way ("resulted from",
1293 32 "is explained by"), "Condition" for "if", "provided that", "only when", and
1294 33 "Instantiation" when one claim is an INSTANCE of the other, i.e. a specific
1295 34 case of a general statement ("for example", "specifically", "such as").
1296 35 - Use "Alternative" only when the two claims cannot both be true AND cannot both
1297 36 be false. Use "Contrast" when they merely cannot both be true - look for "by
1298 37 contrast", "whereas", "on the other hand", or two claims set against each other
1299 38 without either being conceded.
1300 39 - Use "Concession" when the text grants a tension and then overrides it -
1301 40 "although", "despite", "even though", "notwithstanding", or a clause that admits
1302 41 one claim while asserting the other anyway. A concessive marker attaching two
1303 42 claims IS a relation between them, even when the concession sits in a
1304 43 subordinate clause rather than its own sentence.
1305 44 - Use "Disjunction" only when the response rules out both claims being false
1306 45 ("at least one of", "one or both", "perhaps both").
1307 46 - Use "Precedence" or "Succession" only when the ordering carries no truth
1308 47 dependence; set their coupling to "none". These ARE relations and must be
1309 48 emitted: "before", "after", "subsequently", "earlier", "later", or an explicit
1310 49 statement that a sequence carries no causal or inferential force. Read such a
1311 50 pair as "Precedence" when the source comes first in time and "Succession" when
1312 51 it comes second. Do not silently upgrade one to "Cause-Effect" or "Evidence"

```



```

1296 52 because the two events sit next to each other in an argument.
1297 53 - Relate a pair only ONCE unless the response genuinely draws two different
1298 54 connections between them. Emitting many relations does not make the recovery
1299 55 more complete; a pair the response merely mentions in the same paragraph is not
1300 56 related.
1301 57 - "coupling" is not an independent choice: it must be the coupling that the
1302 58 "sense" you chose compiles to. Cause-Effect, Effect-Cause, Evidence,
1303 59 Condition and Instantiation are "entailment"; Restatement is "equivalence";
1304 60 Contrast and Concession are "contradiction"; Alternative is "exclusive";
1305 61 Disjunction is "co_necessity"; Precedence and Succession are "none".
1306 62 - Pairs the response leaves unrelated must be omitted entirely.
1307 63
1308 64 The function returns a JSON list. Note that your response will be passed to the
1309 65 python interpreter, SO NO OTHER WORDS!
1310 66
1311 67 recover_relations(response="[RESPONSE_PLACEHOLDER]", atoms=[ATOMS_PLACEHOLDER])

```

## A.7 V3 — PROSE AUDIT

2,457 characters, 40 lines. Placeholder: [RESPONSE\_PLACEHOLDER]. The extensive list of spans that must *not* be flagged as leakage is the calibration described in §6.

```

1313 1
1314 2 Let's define a function named check_response(response: str).
1315 3 The function audits a generated response and returns a JSON object:
1316 4 { "fluency": 1-5, "formality": 1-5, "organization": 1-5,
1317 5   "leakage": [list of quoted spans], "hedging": [list of quoted spans],
1318 6   "artifacts": [list of quoted spans] }
1319 7
1320 8 - fluency, formality, organization: 5 = indistinguishable from expert human
1321 9 prose; 1 = badly degraded.
1322 10 - leakage: any span revealing the response was generated from an annotation
1323 11 plan. Read this NARROWLY: naming a relation type or sense; naming the plan,
1324 12 the strength bands, or the validity labels; referring to "claims", "atoms" or
1325 13 "statements" as the objects the text was assembled from; or any label, tag,
1326 14 numeric index, or bracketed marker. Do NOT flag a connective or coordination
1327 15 that merely realizes a relation - "either X or Y", "at least one of X or Y",
1328 16 "at least one of these was true", "perhaps both", "one or both",
1329 17 "the two cannot both be true, and one of them must hold",
1330 18 "one of these accounts must be wrong", "although X, Y",
1331 19 "before", "after", "subsequently", "earlier than", "which postdates",
1332 20 "as indicated by", "as shown by", "which indicates", or an
1333 21 explicit statement that an ordering carries no causal or inferential force. A
1334 22 writer is REQUIRED to use those, and a coupling over two claims' joint truth
1335 23 cannot be expressed without one. Every span you report must be QUOTED VERBATIM
1336 24 from the response; if you cannot copy it out of the text, it is not there and
1337 25 must not be reported. Do NOT
1338 26 flag ordinary scholarly vocabulary used about the subject matter, such as
1339 27 "claim", "outcome", "reading", "account" or "finding".
1340 28 - hedging: a span using one of these EXACT words: "assume", "might",
1341 29 "possibly", "allegedly", "supposedly", "reportedly", "it is claimed". The list
1342 30 is CLOSED - quote the word you found, and flag nothing that does not contain
1343 31 one of them. Uncertainty a writer is REQUIRED to express is not hedging: "at
1344 32 least one of X or Y, perhaps both" and "either X or Y" state the logical
1345 33 strength of a relation, not doubt about the content, so do NOT flag them.
1346 34 - artifacts: enumerated list structure, headings, repeated template phrasing, or
1347 35 a sentence that reads as a stitched-together claim rather than prose.
1348 36
1349 37 Return the JSON object only. Note that your response will be passed to the
1350 38 python interpreter, SO NO OTHER WORDS!
1351 39
1352 40 check_response(response="[RESPONSE_PLACEHOLDER]")

```

## A.8 V4 — CLAIM COVERAGE

1,405 characters, 26 lines. Placeholders: [RESPONSE\_PLACEHOLDER], [ATOMS\_PLACEHOLDER].

```

1353 1
1354 2 Let's define a function named check_atom_coverage(response: str,
1355 3 atoms: list[str]).
1356 4 For each atom, the function decides whether the response asserts it, returning a
1357 5 JSON list of objects:
1358 6 { "index": i,

```

```

1350 7 "status": one of ["asserted", "altered", "missing", "merged"],
1351 8 "span": the quoted span of the response that asserts it, or null,
1352 9 "position": the 1-based index of that span among the response's sentences}
1353 10
1353 11 - "asserted": the response asserts the atom's full content, in any wording.
1354 12 - "altered": the response asserts something related but changes the atom's
1354 13 content (a different number, date, entity, or polarity).
1355 14 - "missing": the response does not assert the atom.
1356 15 - "merged": the atom's content has been absorbed into another atom's, so it is no
1357 16 longer separately assertable - you could not negate this atom alone and leave
1357 17 the other standing. Reserve this for genuine loss of content.
1358 18 Two atoms joined by a connective are NOT merged: "either X or Y", "at least one
1359 19 of X or Y", "although X, Y", "X, whereas Y", and "X; that is, Y" each assert
1359 20 BOTH atoms, because each remains independently true or false. Mark both
1360 21 "asserted", with the span covering the part that asserts each one.
1361 22
1361 23 Return the JSON list only. Note that your response will be passed to the python
1362 24 interpreter, SO NO OTHER WORDS!
1362 25
1363 26 check_atom_coverage(response="[RESPONSE_PLACEHOLDER]", atoms=[ATOMS_PLACEHOLDER])

```

## B THE PROMPT SPECIFICATION AND SUBSTITUTION CONTRACT

Each prompt's numbered-instruction count and required placeholder set are declared in a specification table, and a test asserts the declaration against the prompt strings themselves. The mechanism matters because the alternative failure is silent: a corpus generated by prompts that no document describes.

| Prompt | Instructions | Required placeholders    |
|--------|--------------|--------------------------|
| P1     | 9            | TOPIC                    |
| P2     | 7            | QUESTION                 |
| P3     | 10           | CLAIMS, QUESTION         |
| P4     | 8            | PLAN, QUESTION           |
| P5     | 5            | OPERATOR, PLAN, RESPONSE |
| V1     | 0            | ATOMS, RESPONSE          |
| V3     | 0            | RESPONSE                 |
| V4     | 0            | ATOMS, RESPONSE          |

Instructions are counted by matching lines that begin at column one with a digit followed by a period and a space, so indented sub-items — P3's per-sense glosses, P5's per-call definitions, P2's lettered composition breakdown — do not inflate the count. Placeholders are extracted by matching `[A-Z_] + _PLACEHOLDER` in brackets, deduplicated and sorted.

Substitution enforces a *bidirectional* exact match between the placeholders a prompt declares and the values supplied: a missing name and an extra name are both errors. The reason is that a typo in either direction otherwise ships a prompt containing a literal `[FOO_PLACEHOLDER]`, which a model will dutifully treat as content — a failure that produces plausible output and no error.

**A validator that was specified and removed.** The original design included a fourth validator to adjudicate exhaustiveness, deciding whether an opposition was EXCLUSIVE or merely CONTRADICTION. It is deliberately absent. Those couplings are derived from the relation sense by the compile map of Table 1, which every other part of the harness treats as authoritative, so the validator would have re-derived from prose what the taxonomy fixes by construction — and, worse, could have contradicted it. Exhaustiveness is instead a property of the sense the planner chose and the writer was required to realize explicitly, checked by V1's sense-recovery rate rather than adjudicated separately.